

DISTRIBUTED COMPUTING

Subject Code: 25MTCSEDE52

Hours: 45

L:T:P:E: 3:0:0:2 Total

Credits: 3 Semester: I

Course Objectives:

1. To understand the core principles, characteristics, and operational frameworks of distributed systems.
2. To analyze fault tolerance, consistency, and security mechanisms in distributed environments.
3. To explore virtualization and cloud computing technologies, focusing on service delivery, deployment models, and advanced architectures.

Course Outcomes (COs):

By the end of the course, students will be able to:

Course Outcomes	Description	Bloom's Taxonomy Level
CO1	Analyze the foundational concepts, communication mechanisms, and process synchronization techniques in distributed systems	L4
CO2	Examine fault tolerance approaches, consistency models, and security mechanisms to enhance distributed system reliability	L4
CO3	Compare virtualization strategies and resource management techniques, including virtual networks and machine models	L4
CO4	Evaluate different cloud service delivery and deployment models for their applicability in real-world scenarios	L4
CO5	Analyze the features and scalability of advanced cloud architectures to address complex computing requirements	L4

Module 1: Introduction to Distributed Systems

Introduction: Definition, Characteristics, Hardware and Software Concepts, Distributed Operating System, Network Operating System

Communication and Synchronization: Layered Protocols, TCP/IP Protocol Suit, Client-Server Model, Remote Procedure Call, Processes, Threads, Mutual Exclusion, Deadlocks

Module 2: Fault Tolerance and Security in Distributed Systems

Fault Classification: Crash Faults, Byzantine Faults

Fault Tolerance Approaches: Proactive- Self Healing, Pre-emptive migration, System Rejuvenation;
Reactive- Checkpoint Restart, Job Migration, Replication
Consistency: Consistency Spectrum, CAP Theorem, Consistency Models in Practice
Security: CIA Triad, Threats, Authentication, Authorization, Encryption, Secure Communication Protocols,
Access Control, Trust Management, Privacy Preserving, Auditing and Monitoring

Module 3: Virtualization and Cloud Computing

Virtualization: Basic Concept, Hardware Abstraction Layer, Virtual Machine Monitor
Resource Management: CPU Management, Memory Management, Storage Management
Virtual Networks: Virtual LAN, Virtual Private Network, Software Defined Network, Network Function
Virtualization, Overlay Networks
Information Model for Virtual Machines: Resource Model, State Model, Allocation Model, Template
Model,
Performance Model, Security Model, Lifecycle Management Model, Data Persistence Model, Migration
Models Introduction to Cloud Computing: Definition, Characteristics, Mathematical Model

Module 4: Cloud Service and Cloud Computing

Service Delivery Models: Infrastructure as a Service, Platform as a Service, Software as a Service,
Everything as a Service
Service Deployment Models: Public Cloud, Private Cloud, Hybrid Cloud, Community Cloud

Module 5: Cloud Computing Architecture

Fundamental Cloud Architectures: Workload Distribution, Resource Pooling, Dynamic Scalability, Elastic
Resource Capacity, Service Load Balancing, Cloud Bursting, Elastic Disk Provisioning, Redundant Storage
Advanced Cloud Architectures: Hypervisor Clustering, Load Balanced Virtual Server Instances, Non-
Disruptive Service Relocation, Zero Downtime, Cloud Balancing, Resource Reservation, Dynamic Failure
Detection and Recovery, Bare-Metal Provisioning, Rapid Provisioning, Storage Workload Management
Specialized Cloud Architectures: Direct I/O Access, Direct LUN Access, Dynamic Data Normalization,
Elastic Network Capacity, Cross-Storage Device Vertical Tiering, Intra-Storage Device Vertical Data
Tiering, Load Balanced Virtual Switches, Multipath Resource Access, Persistent Virtual Network
Configuration, Redundant Physical Connection for Virtual Servers, Storage Maintenance Window

Text Books:

1. George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair – Distributed Systems: Concepts and Design, 5th Edition, Pearson Education, 2011.
2. Andrew S. Tanenbaum and Maarten Van Steen – Distributed Systems: Principles and Paradigms, 2nd Edition, Pearson Education, 2006.
3. Kai Hwang, Jack Dongarra, and Geoffrey Fox – Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Morgan Kaufmann, 2012.

Reference Books:

1. Ajay D. Kshemkalyani and Mukesh Singhal – Distributed Computing: Principles, Algorithms, and Systems, Cambridge University Press, 2008.
2. Pradeep K. Sinha – Distributed Operating Systems: Concepts and Design, PHI Learning, 1997.
3. Rajkumar Buyya, Christian Vecchiola, and Thamarai Selvi – Mastering Cloud Computing: Foundations and Applications Programming, Morgan Kaufmann, 2013.

4. Randy Chow and Theodore Johnson – Distributed Operating Systems and Algorithm Analysis, Pearson Education, 1997.
5. Thomas Erl, Zaigham Mahmood, and Ricardo Puttini – Cloud Computing: Concepts, Technology, and Architecture, Prentice Hall, 2013.

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline, paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell